

ESTIMATING OF SITE OVERHEADS FOR RESIDENTIAL PROJECTS IN EGYPT USING NEURAL NETWORK

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ABSTRACT

Real estate industry has continued to prosper over the past 7 years in Egypt despite the hard-economic conditions, and the currency liberalization followed by a high inflation rate. As such, construction companies and contractors need to enhance the accuracy of the budget to achieve the targeted profit. The budget cost includes two types of cost, which is the direct cost and the general costs. The last one composed of two types of costs as Home overheads and site overheads. The home overheads are not associated to a certain project, site overheads are the expenses associated to a certain project but not to a certain work package. The main objective of this research is to establish a neural network model for residential projects to predict the percentage of the site overheads of the direct cost. This will contribute to enhance the estimate process and budget. Information related to main factors, and its weights affecting site overhead percentage were identified by experts through a structured questionnaire based on calculated sample size. In addition, cost data for 55 different projects in the past seven years was collected with different conditions of company class, direct cost, project duration, project location, contract type, and type of company ownership. These data were used as data base for the learning process of the neural network. The best model developed consists of six input neurons, two hidden layers with six and five neurons respectively, and one output layer representing the site overhead percentage.

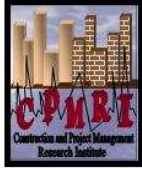
Keywords: Residential Projects, Egypt, Cost Estimate, Artificial Intelligence, Site Overheads, Artificial Neural Network.

INTRODUCTION

Construction industry is one of the major sectors in the Egyptian economy, especially real estate industry. Construction and building sector has a great impact on the Gross Domestic Product (GDP). However, after the Egyptian revolution in 2011 and the currency liberalization in November 2016 followed by high inflation rate, many sectors suffered due to the unstable economy situation accompanied by the political risks in Egypt.

With all these factors into consideration, an accurate estimate for the project cost in highly needed by building construction companies in Egypt, and controlling project costs has become more important and with greater impact than before.

Cost estimate is not only used for preparing the project budget, but also serves other purposes. It is a tool for project cost control and to better monitor the project cost performance. The term "Cost Engineering" is the relation between estimating, controlling, and scheduling. These are the drivers for a successful and cost-effective project. Thus, an accurate cost estimate is essential for the success of a project and to maximize the profit of the construction company.



Most of the building construction companies find no hassle in estimating the direct cost of a project, the inaccuracy appears in the estimating of the overhead costs, causing a clear cost variance between estimated budget and actual cost either cost over-run or a cost saving. Both cases can cause a problem for the construction company.

The absence of structured and accurate methods that can assess site overhead cost for residential projects in Egypt put construction companies in a risk of inaccurate estimate of bid package that may affect the profit margin of the company.

LITERATURE REVIEW

Overhead costs:

Cost estimating is an essential element in construction business as supported by many researches. Carty stated that it is a key function in order to enable companies acquiring new contracts and projects at the accurate price so as to survive in the market. [1]

In construction industry, the contractor is obligated to offer a price to the client before the construction process begins. As opposed to manufacturing industry where the product price can be determined after the final product is finished and produced. [2]

Construction Company overhead is commonly categorized into 2 branches: Site (Project) overhead costs, and General (Home) overhead costs. [3]

Site overhead costs include costs that cannot be allocated on a certain work package, but it is essential for activities to be completed, in other words, it serves the execution process. General overheads are generally considered as costs of conducting business, these are costs that cannot be allocated on a certain project. [4]

A study done by Nancy Holland and Dana Hobson previously identified overhead costs the same as Dagstino and Feigenbaum did. Also set examples for what may site overhead costs cover like, site staff, cleaning activities, temporary services, first aid and safety provisions, transportation of site operatives, and site accommodation. [5-4]

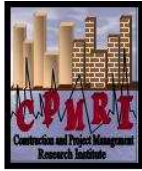
It is difficult for construction companies to estimate the optimum percentage for overhead costs, especially through the unstable environment and construction market that enable them to acquire a tender and maximize their profit as well. If the overhead costs are not accurately estimated prior bid submission, this may lead to a cost overrun and the attempts to recover this overrun can result in a financial issue and collapse for the construction company. [6]

Neural Networks versus other programming

Neural networks differ from the conventional programming approach in the way they recognize and solve the problem [7]. The conventional programming algorithm involves a set of rules or instructions to be followed when countered by a certain problem, in order to get to the solution. These rules or instructions are translated into signals in which the computer program can follow to reach the solution. Therefore, the computer recognition of the problem and its solution is limited to the programmer's understanding and the way of solving it. This is the main difference between the conventional programming and the neural network that the conventional approach cannot solve the problems that the programmer does not understand. [7]

However; conventional programming and neural networks are not in competition with each other, but they complete and work with each other. There are tasks that are more suitable to conventional programming, and others that are suitable for neural networks. Moreover; there are tasks that require for both algorithms to work together in which the conventional algorithm supervise the neural network in order to acquire maximum efficiency. [8]

And unlike fuzzy logic that allows making decisions based on imprecise data by developing membership functions and rules to relate them, ANNs includes a learning process that involves learning algorithms and requires training data. Fuzzy logic is suitable for inaccurate data set; on the other hand ANNs are more practical if the data set for the training is accurate and relatively enough to train the network. [9]



Previous work on Neural Networks in Cost Estimation

The behavior of a project lifecycle and its trends can be forecasted before the actual beginning of the project through simulations and modeling. [10]

Using of artificial intelligence like expert systems and neural networks has proven to be helpful in solving prediction problems. [11]

In Egypt, Georgy and Barsoum developed an ANN model for the parametric cost estimate of schools construction projects. They used statistical and neural network models for estimating costs; the research finding was a neural network of a single hidden layer with number of neurons equal to two thirds the number of neurons in the input layer would be sufficient. [12]

Hosny used the AAN modeling to predict the cost increase of construction projects in Egypt based on the analysis of several factors influencing project cost such as project type, contract type, owner behavior, design completion. [13]

In Turkey, Gunaydin and Dogan designed an Artificial Neural Network model to predict the cost per meter square for reinforced concrete structural systems of buildings in an early phase of the design process. The used data for developing the network consists of 30 construction projects varies from four to eight stories residential buildings. The developed ANN architecture consisted of eight input layer (one for each input variable). The resulted model had an accuracy of 93% in estimating the cost per meter square for reinforced concrete structural residential buildings. [14] Arafa & Alqedra highlighted that researches on using neural networks in the field of cost estimation is very limited. In spite of the fact that there are many researches that discussed neural networks approach in different engineering fields. [15]

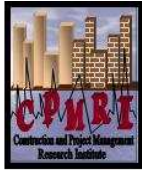
In Gaza Strip, Arafa & Alqedra developed an Artificial Neural Network model to predict construction cost at an early project phase. The used data base for learning consisted of 71 construction project from Gaza Strip. The resulted ANN model architecture consisted of one hidden layer with seven neurons. The established model has proven to be relatively successful in predicting construction cost at an early stage, and without the need of detailed information about the project. [15]

In Philippines, Lyne and Maximinio developed an AAN to estimate the total structural cost of building construction projects in Philippines based on six parameters which are: number of floors, number of basements, area of floor, concrete volume, area of formwork, and weight of reinforcement steel. The data used for the learning process consists of data from 30 different building construction projects; this data set was divided into 60%, 20%, and 20% for training, validation, and testing respectively. The resulted model architecture was a network of six input neurons (one for each input parameter), one hidden layer with seven neurons, and one output neuron representing the structural cost. The model was developed using MatLab with a feed forward back propagation technique. [16]

RESEARCH METHODOLOGY

To develop the neural network to predict the site overhead percentage, this research has to go through different phases in order for this neural network to accurate and reliable. The extensive literature study in the previous section serves in identification of the factors that may influence the percentage of site overhead. These factors mainly include project location, direct cost, duration, complexity, payment schedule, contract type, tendering method, class of company, contractor cash availability, firm need of work, and number of competitors. A survey was conducted to focus on the factors that influence the site overhead percentage in the Egyptian industry.

The next step was to collect project data from the past 7 years for the aim of learning the neural network from actual project data. A number of 55 Egyptian residential projects data was collected from various number of construction companies in Egypt. A comparative analysis was then performed on these project data in order to illustrate the influence of each factor on the percentage of site overhead.



Then the process of designing the network is ready and the main objective of the research; which is developing a network for estimating the site overhead percentage.

After reaching the optimum network structure, it is critical to test and validate the network, 6 projects were set aside for the validation process to be performed.

DATA COLLECTION AND ANALYSIS

The process of data collection is divided into two stages: the first stage is identify the factors influencing the site overhead costs in the Egyptian residential construction industry and comparing the factors resulted from the literature review with those affecting the construction industry in Egypt.

The second stage is to collect data for different residential construction projects in Egypt, with the data and factors resulted from the analysis of the first stage of the data collection and analysis.

Site overhead factors questionnaire

This questionnaire is designed to investigate the factors affecting site overhead percentage in the residential building construction projects in Egypt. And participants were selected to have an experience more than 10 years in residential construction market in Egypt working in estimation, tendering, control, financial and commercial technical office section, procurement, and others.

The first section of the questionnaire consists of thirteen multiple choice questions, where participants were asked to rate each factor according to their influence on the site overhead percentage and how each factor can affect this percentage. The participants chosen are experts in the residential construction industry from the pioneer companies in the field of residential construction in Egypt.

The second part of the questionnaire is where the participant was asked if there is any other factor that was not stated in the first section that may affect site overhead percentage in the Egyptian residential construction.

The results of the questionnaire as illustrated in Fig. 1 shows that the top five factors that most affect the site overhead percentage in residential construction in Egypt are:

- Project Location
- Project Direct Cost
- Project Duration
- Contract Type
- Class of Company

And the most stated factor mentioned in the second section of the questionnaire is whether the company is privately or publically owned.

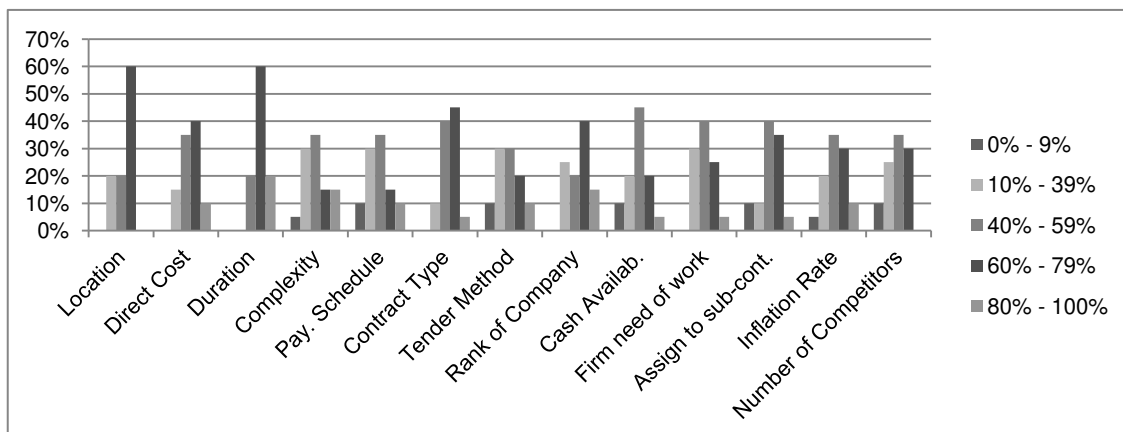
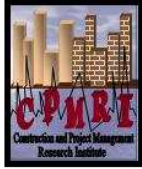


Fig. 1: Questionnaire (1) results



Major Factors Analysis

The results from the previous questionnaire were categorized in as follow:

- Class of Contracting Company: First, Second, Third, and Forth Class
- Project Direct Cost in Million EGP: Less than 50, from 50 to 150, from 150 to 300, from 300 to 450, and more than 450
- Project Duration in Months: less than 18, from 18 to 36, from 36 to 60, and more than 60
- Project Location: Inside the capital city, and Outside the capital city
- Contract Type: Cost Plus, Unit rate, and lump-sum
- Ownership of company: Private, and Public

At this phase of the data analysis, a second questionnaire was conducted and experts from residential construction industry were asked to weight the previous six factors and the weight of each sub-category. In order to understand the influence of input for the factors governing the site overhead cost percentage.

Experts are selected according to their experience in the Egyptian residential construction market to be more than 20 years working in residential projects managerial positions or cost related positions.

Table 1 presents the weights for each factor affecting site overhead percentage resulted from the questionnaire.

Table 1: Weights resulted from questionnaire (2)

Factor	Weight
Rank of Company	19%
Direct Cost	12%
Project Duration	19%
Project Location	15%
Contract Type	25%
Ownership Type	10%

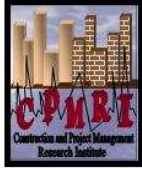
Projects data for site overhead percentage

Data from 55 different projects in the past 7 years were gathered in order to be used in the learning process of the network. The data collected for projects was according to the 6 major factors resulted from questionnaire (1). These projects represent residential projects from first, second, third, and forth class contracting companies in Egypt.

A comparative analysis was performed on the collected project data as shown in Fig. 2 to figure out the bounding relationship for each factor and the site overhead percentage.

From the project data analysis, it is concluded that:

- Second category of construction companies has the highest average site overhead percentage. Through extensive research and interviews with industry experts, this results from the strict control system of the first category companies, unlike the second category with a relatively weaker control system.
- Projects with duration between 36 and 60 months have the highest average site overhead percentage. Because with project duration more than 60 months, some of the



- facilities and site overhead influencers may be considered to be under the long term category and no longer will be temporary that results in decreasing their cost.
- Projects with a direct cost between 300 and 450M EGP are projects with highest site overhead percentage, not projects with a direct cost more than 450M EGP. This trend is caused by the relativity between the direct cost of the project and the site overhead percentage. With a very high direct cost, the site overhead tends to be lowered as a percentage from the direct cost, but it will remain high as a monetary value.
 - It is clear that projects outside the capital city tend to have a higher site overhead percentage than those inside the capital city. This has many influencing factor such as the over cost of delivering materials, high accommodation rates, higher fuel consumption, and many other factors.
 - Contract type is a critical factor in determining the overall pricing of the project, and by default the site overhead percentage. In lump-sum contract projects, contractors tend to higher the site overhead percentage in order to lower the risk of cost overrun accompanied by this type of contracts.
 - Contracting companies with a private ownership type tend to utilize all their available resources in order to minimize the expenditures. On the other hand, public sector companies do not the same importance for such factor, so the site overhead percentage tends to be higher in the public sector contracting companies.

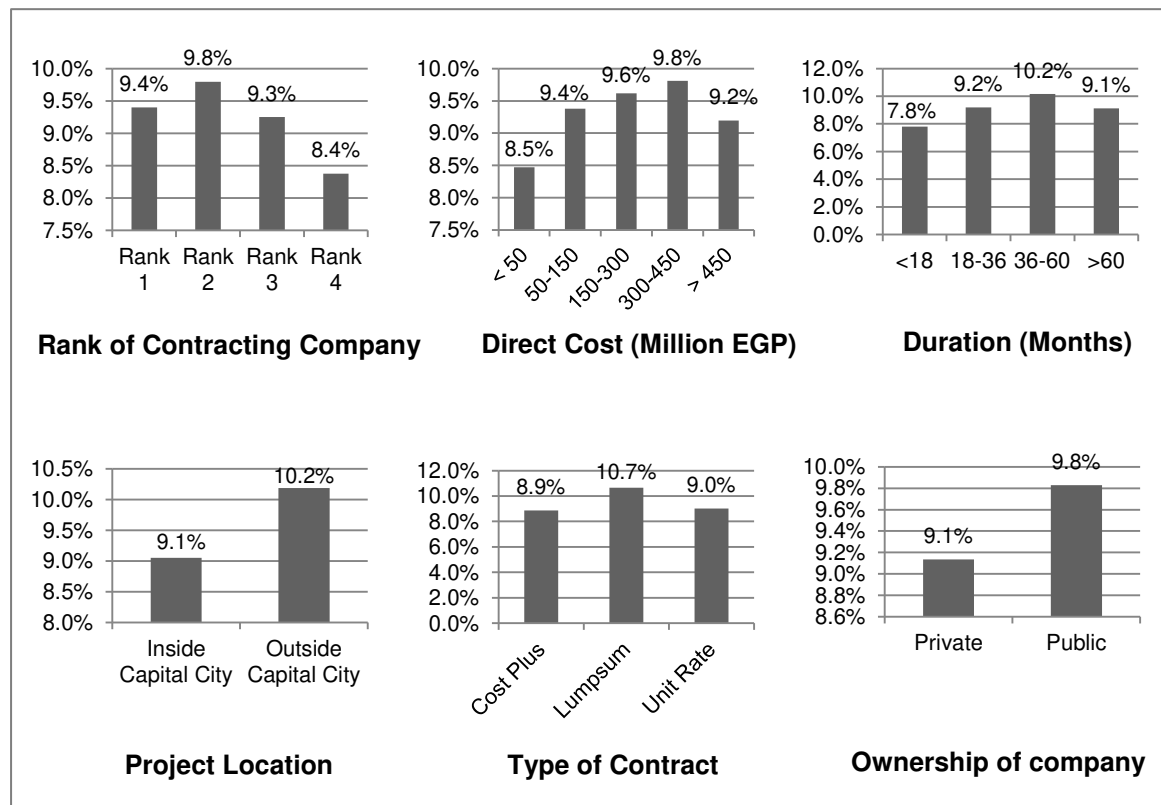
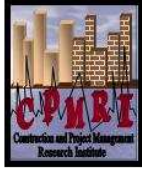


Fig. 2: Comparative Analysis for each Site overhead Factor

The aforementioned data analysis and results set the bases for designing the neural network required for estimating and assessment of the site overhead percentage for residential construction companies in Egypt.



DESIGN OF THE NEURAL NETWORK MODEL

After the data collection and analysis phase is completed, the resulted data is ready for developing and implementing the neural network.

"Neural Designer" software is used for developing the neural network for estimating site overhead as a percentage from the direct cost of the project. The "Neural Designer" is very easy to use and has a friendly user interface to work on. The software is applicable on windows and free to download.

The developing of the model was done using the trial and error method in order to reach the optimum model structure.

This section presents the steps taken to develop the artificial neural network.

1. Collecting the needed project data for the learning process
2. Coding of each input
3. Design the network architecture and choosing the transfer function
4. Determine the learning algorithm
5. Training of the network
6. Validation of the network
7. Testing the network

Data Set

The projects collected for the learning of the network was based on a calculated sample size according to the number of contractors enrolled in the "The Egyptian Federation for Construction and Building Contractors". The calculated sample size was to ensure that the collected project data are reflecting a representing sample.

"Neural Designer" enables the importing of an excel data file. It divides the data set to 60% for learning, 20% for validation, and 20% for testing and verification the network.

Before importing the data set file, project data was coded in order to be properly imported to the "Neural Designer". Fig. 3

Data source


 Import data file


 Import data base

Data file name: ../Coded Project Data.xlsx

Data file preview:

	1	2	3	4	5	6	7
1	Rank of Con...	Direct Cost ...	Project Dur...	Project Loc...	Type of Con...	Ownership ...	OHP
2	2	4	4	1	2	1	9.64
3	2	5	3	1	2	1	9.22000000...
...	...	...	...	...	...	...	...
55	1	1	4	1	2	1	9.43

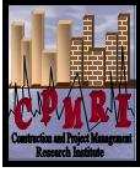
Number of columns:

7

Number of rows:

55

Fig. 3: Data set interface



After importing the data set file, the software categorizes by default all data variables to inputs and outputs, so in the model for estimating site overhead percentage, the six input variables are the rank of company, project direct cost, project duration, project location, contract type, and ownership type of the company. And the output or target variable is the site overhead percentage. Fig. 4

Variables

Default All input All target All unused

	Name	Units	Description	Use
1	Rank of Contracting Co...			Input
2	Direct Cost (EGP)			Input
3	Project Duration (Months)			Input
4	Project Location			Input
5	Type of Contract			Input
6	Ownership Type of Contr...			Input
7	OHP			Target

Number of variables: 7

Input variables: 6

Target variables: 1

Unused variables: 0

Fig. 4: Variables interface

Training of the Network

The most important principle of developing a neural network model is to define which training algorithm to use. Training the network may be done by various learning methods to adjust weights. All the models developed in order to reach an accurate and reliable network model were trained in a supervised learning rule by a back-propagation learning algorithm. In supervised learning, the learning rule is provided with a set of examples (the training set) of proper network behavior where each input to the network corresponds to a correct output.

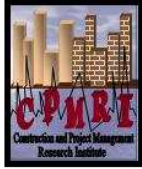
As the inputs are applied to the network, the network outputs are compared to the targets. [17] The learning rule is then used to adjust the weight and biases of the network in order to move the network outputs closer to the targets.

The back-propagation works on the gradual error reduction between inputs and the target output. It develops the inputs to output leveling by minimizing the root mean squared error, and the training process shall be finished when the RMSE become constant

$$RMSE = \frac{\sqrt{\sum_{i=1}^n (xi - E(i))^2}}{n}$$

- Where n is the number of projects to be evaluated in the training phase
- Xi is the model output related to the sample
- E is the target output

The root mean square error is a good overall measure whether a training run was successful.



Architecture of the Model

The "Neural Designer" at this stage needs to set:

- The number of hidden layers
- The number of neurons in each layer
- The transfer function for each layer

The process of designing the network in order to reach the best possible model was done using a traditional trial and error method. The best model was chosen according to the lowest root mean square error and the correspondence error values.

The trial and error of models resulted in a number of 92 models categorized as follows:

- 1 hidden layer with a hyperbolic tangent transfer function. (Fig. 5)
- 1 hidden layer with a sigmoid transfer function. (Fig. 6)
- 2 hidden layers with a hyperbolic tangent transfer function for each layer. (Fig. 7)
- 2 hidden layers with a sigmoid transfer function for each layer. (Fig. 8)

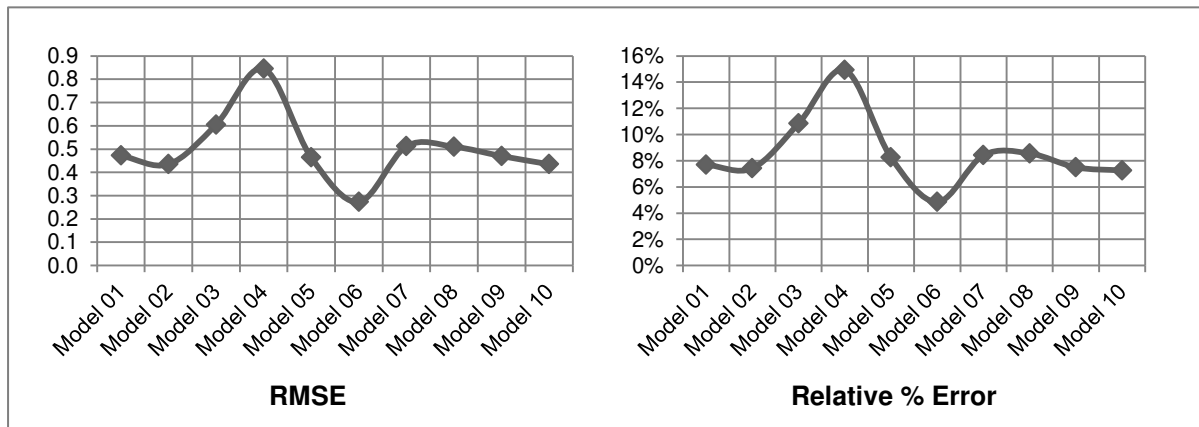


Fig. 5: Model Trials with 1 Hidden Layer and Hyperbolic Tangent function

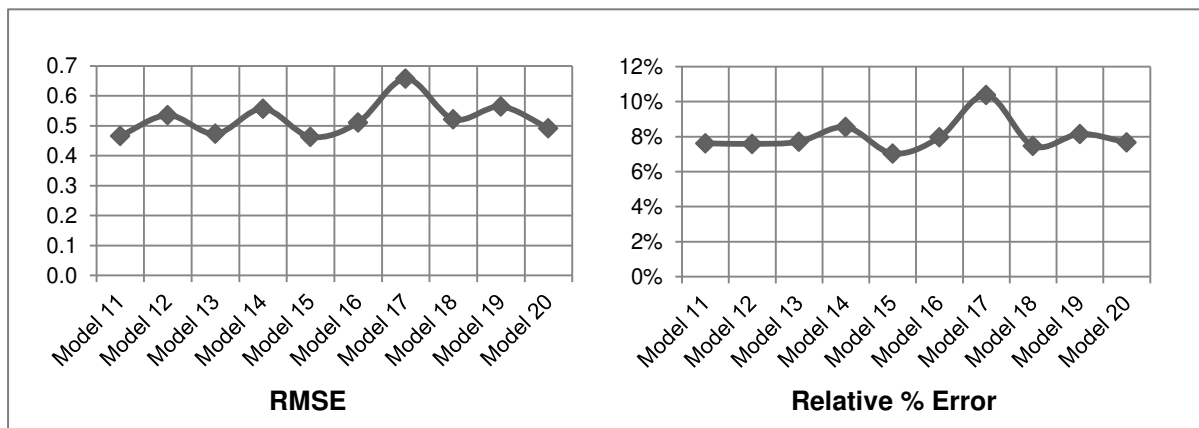
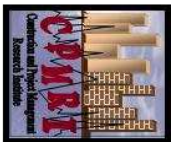


Fig. 6: Model Trials with 1 Hidden Layer and Sigmoid function

As illustrated in the previous fig. (5) and fig. (6), models with 1 hidden layer varies as the number of neurons changes. For 1 layer models with hyperbolic tangent function, RMSE ranges from 0.273 to 0.845, and relative % error ranges from 4.86% to 14.93%. As for the 1 layer models with sigmoid function, RMSE ranges from 0.463 to 0.657, and relative % error ranges from 7.04% to 10.38%.



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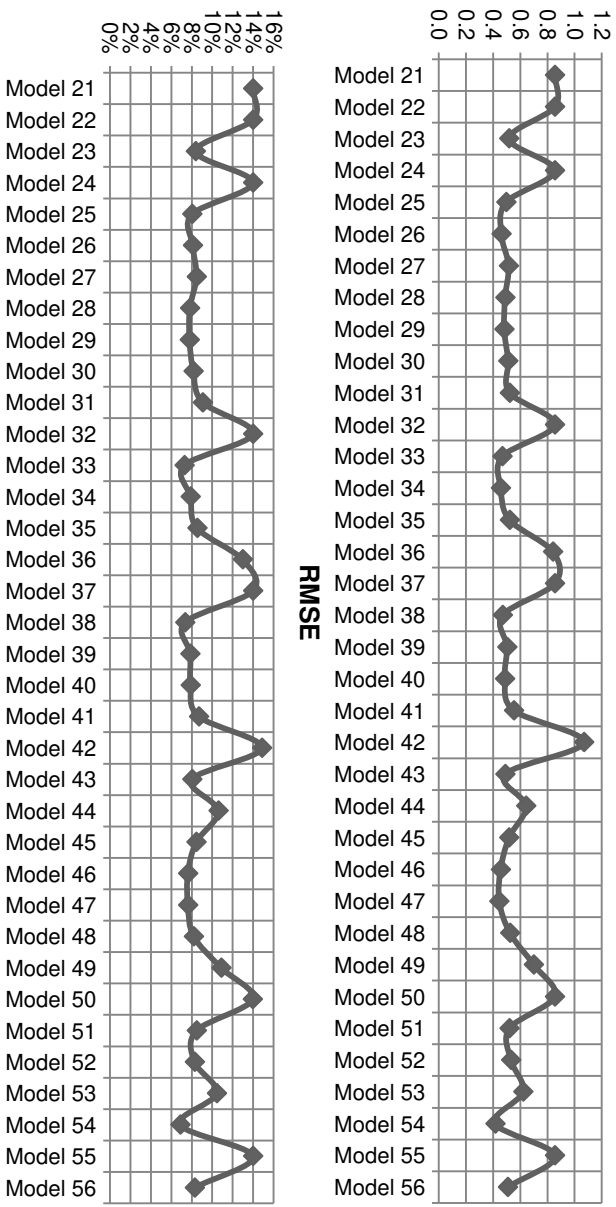


Fig. 7: Model Trials with 2 Hidden Layers and Hyperbolic Tangent function

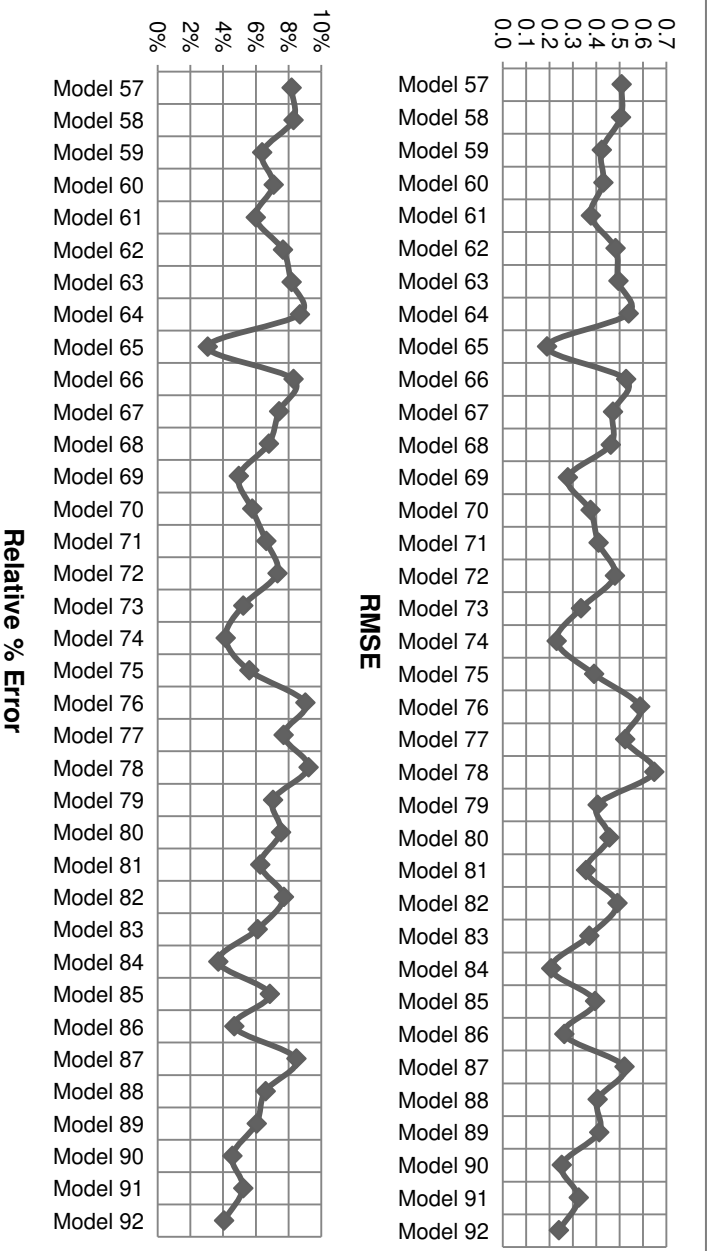
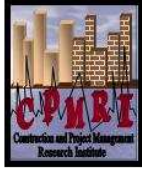


Fig. 8: Model Trials with 2 Hidden Layers and Sigmoid function



As illustrated in Fig. (7) and Fig. (8), models with 2 hidden layers vary as the number of neurons changes. For 2 layer models with hyperbolic tangent function, RMSE ranges from 0.415 to 1.068, and relative % error ranges from 6.88% to 14.84%. As for the 2 layer models with sigmoid function, RMSE ranges from 0.188 to 0.647, and relative % error ranges from 3.05% to 9.21%

The results of the trial models show that Model 65 is the optimum choice with the following design considerations in Table (2) and network architecture as shown in Fig. (9).

Table (2): Model 65 design considerations

No. of input neurons	No. of Hidden Layers	No. of neurons in hidden layer 1	No. of neurons in hidden layer 2	Transfer function	RMSE	Relative % Error
6	2	6	5	Sigmoid	0.188	3.05%

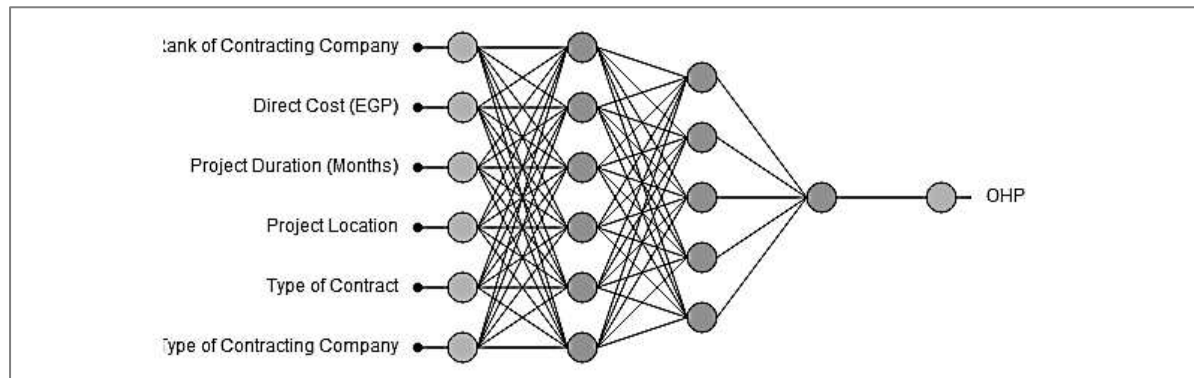


Fig. (9): Model 65 neural architecture

Testing of the Model

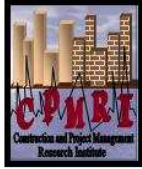
To test and evaluate the predictive model and its accuracy, 6 projects were randomly selected aside from the previous 55 projects used for the learning of the network. The relative error between the predicted and the real life values determines the accuracy of the model. Table (3)

$$\text{Relative \% Error} = \frac{(\text{Real life value} - \text{Predicted value})}{\text{Real life value}} \times 100$$

Table (3): Model 65 Assessment

Project Number	Model Prediction	Real life Value	Relative Error	Assessment
Project 1	10.648	10.590	-0.548%	Accepted
Project 2	9.878	10.210	3.252%	Failed
Project 3	10.193	9.980	-2.134%	Accepted
Project 4	8.588	8.790	2.298%	Accepted
Project 5	7.633	7.550	-1.099%	Accepted
Project 6	8.604	8.750	1.669%	Accepted

The model predicted 5 out of 6 projects within the accepted relative error, which represents 83.33% accuracy. This percentage considered to very accurate and accepted for the residential construction industry in Egypt.



CONCLUSION

The study demonstrates the benefits of using the neural network technology in the estimating process of the site overhead costs, and the undeniable prediction advantages are its simplicity, calculation speed, and accuracy. Results of testing the proposed model proved to be very accurate by 83.33%. To sum up, the model is much more accurate and simple to use. That will enhance the contractor's ability to accurately predict their costs, and so maximize their profit. It is recommended for construction companies to consider using of artificial intelligence and to establish a database for the finished and on-hand projects to be used as historical data and benefit from the lesson learned.

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